US Divisional Personnel Data

INTRODUCTION

Daily personnel data for US armored, infantry, and airborne divisions is derived from various primary source records, including the 12th Army Group G-1 Daily Summary, G-1 records of divisions themselves, and G-1 records of selected corps to which the divisions belonged. Personnel data includes daily on-hand strengths in total and by MOS categories (for definitions of the so-called MOS categories, see the UNIT DATA BASE DEFINITIONS section of the ACSDB Unit Data Base narrative), daily attrition (killed, wounded, captured/missing, and DNBI -- disease and non-battle injury), and daily replacements and returns to duty (RTDs) in total and by MOS categories.

Compilation of personnel data for US divisions in the Ardennes Campaign Simulation Data Base (ACSDB) proved to be a complicated task, despite the abundance of primary source records. Obstacles encountered in the task resulted from contradictory or incomplete data in the records, and the requirement to make a daily accounting of on-hand, replacement, and RTD personnel strengths by MOS categories.

This narrative serves as the bibliographic reference for the sources used in compiling US divisional personnel data. It also explains any data estimation, interpolation, or extrapolation employed in generating the data. The narrative should be examined and consulted with other US Unit Data Base narratives for a complete summary and explanation of the personnel data recorded for US divisions in the Unit Data Base. It is arranged by division, with infantry divisions first, followed by armored divisions, and finally the three airborne divisions which participated in the Ardennes Campaign.

One source frequently referenced in this narrative is the US 12th Army Group G-1 Daily Summary (hereafter cited in this report as "12th AG G-1" or "12th AG G-1s"). This document was obtained from the US National Archives in Record Group 407, Box 1753, File 3 and Box 1754, File 1. It contains daily personnel strengths, battle casualties, DNBI, and "reinforcements" for divisions, corps and army combat troops, and corps and army service support troops of the US 12th Army Group. Its value to generation of personnel data in the ACSDB is great, although it has some limitations, as discussed below.

Most sources used in the generation of US personnel data are

from the US National Archives Washington National Records Center in Suitland, Maryland. Unless otherwise indicated, they are from Record Group (RG) 407. They are identified in the narrative by their box numbers, the Archives record identification code, and a brief description of the record including, when appropriate, dates. Some other sources are from microfilm records at the Office of Air Force History, Bolling Air Force Base, Washington, D.C. These are identified with their roll number (a four-digit number preceded by the letter "C") and their classification number (with a decimal), followed by a brief description of the item, i.e., "C5141, 585.034 to 585.04A, 4th ID records including 'Reinforcements for the Month of January 1945.'"

Attached to the end of this paper are a series of charts and tables showing the data used for derivation of various averages. Most of these averages relate to the personnel MOS categories. One of the requirements of the ACSDB is to provide daily on-hand strengths of personnel by MOS, as well as daily replacements and RTDs by MOS. Generation of the MOS data is achieved for most US divisions through the use of computer programs. Computer programs are utilized to save lengthy and cumbersome calculations with hand calculators. The programs calculate daily replacement and RTD MOS strengths using averages, and then calculate daily on-hand MOS strengths using the calculated replacement/RTD MOS strengths and MOS attrition percentages. This process does not compromise the accuracy of the MOS data, as it is simply not available on a daily basis for on-hand strengths, casualties, and reinforcements in primary source records of most divisions. For those divisions for which it is available on a daily basis, such as the 30th Infantry Division, it is incorporated in the ACSDB.

The 12th Army Group G-1s mentioned above proved to be a valuable source of personnel data used in the ACSDB. They contain in one source personnel data of all divisions in the US 12th Army Group for the period 15 December 1944-16 January 1945. This data is as of 2400 hours each day, and this fact, coupled with the 12th AG G-1s' extensive use as a source of data, is probably the single most important reason that US personnel data is as of end of day in the ACSDB. The 12th AG G-1s provide total on-hand personnel strengths, casualties by killed, wounded, captured/missing, and DNBI, and (in some cases) reinforcements on a daily basis, as well as cumulative casualties by killed, wounded, etc. These cumulative casualties proved to be very useful in analysis of divisional personnel data and are mentioned frequently in the discussions on the individual divisions.

The 12th AG G-1s are not without their limitations. Principal among these is their failure to record any daily reinforcement data for units of the US First Army. Another limitation is the lack of differentiation between replacements and returns to duty (RTDs) for divisions of the US Third and Ninth Armies. All reinforcement data is recorded under one

heading in the 12th AG G-1. Upon examination of this data and its comparison with that found in corps and divisional personnel records, it was determined that the data may be either replacement data only or combined replacement/RTD data.

Perhaps the most serious limitation of the 12th AG G-1s, and one which upon deliberation should not be unexpected, becomes evident only upon analysis of the "RTD Tracking Program Printouts" found in an attachment at the end of this paper. These printouts are generated by a computer program which takes the recorded strength of a unit and on a daily basis adds to and subtracts from it the gains (reinforcements -- replacements/RTDs) and losses (battle and non-battle casualties) of the unit. A calculated strength based on this mathematics is provided and the difference, if any, between the calculated strength and the next day's recorded strength is shown as a positive or negative number. Not surprisingly, there are numerous differences (or deltas, as they are called throughout this paper), between the calculated and actual recorded strengths. This is not surprising when one considers that the 12th AG G-1s were compiled <u>on a daily basis</u> with information sent up to the army group from divisions, information which was frequently only estimated through a process which no doubt lacked accuracy during the severe and confused combat of the Ardennes Campaign's initial stages. Other reasons for deltas become manifest upon examination of the RTD Tracking Program Printouts. Occasionally, divisional personnel data leaves out personnel strengths of organic divisional units or, conversely, includes the personnel strengths of non-divisional formations, a phenomenon best exemplified by the 12th AG G-1 data of the US 3d and 10th Armored Divisions. If no reinforcement data is provided for a division, a series of deltas with positive values generally appear in the printout. Often, the reason for the deltas can only be determined by scrutinizing the increases or decreases in the cumulative casualties, caused by corrections made to casualty reports and incorporated in the 12th AG G-1s after the fact. In some cases no explanation can be found for the deltas, as is the case with the 35th ID.

Despite these limitations, the utility of the 12th AG G-1s for the ACSDB is great, and extends far beyond the convenience of having much data recorded in one source, This utility becomes most evident when the personnel data is analyzed in the format of the RTD Tracking Program Printouts. Presented in this manner, the data proves most useful in estimating reinforcements for divisions lacking this data. Attachment 9 to this paper contains both original and final RTD Tracking Printouts for selected divisions, the former containing unmodified data and the latter showing data as it is recorded in the ACSDB. These printouts should be used in the review of the individual division narratives which follow.

It is recommended that the attachments to this paper be read

prior to the review of the individual divisional narratives. Readers will then be able to familiarize themselves with the processes used in generating US divisional personnel data. A profusion of terms and references unique to the ACSDB are used in the divisional narratives, and, unless familiar to the reader, these may seem confusing.

US Non-Divisional Personnel Data

INTRODUCTION

This paper serves as the bibliographic reference for sources used in generating personnel data for US non-divisional units as recorded in the Ardennes Campaign Simulation Data Base (ACSDB), to include daily on-hand strengths in total and by MOS categories, daily attrition (killed, wounded, captured/missing, and disease and non-battle injury (DNBI)) and daily replacements and returns to duty (RTDs) in total and by so-called military occupational specialties (MOS) categories. For definitions of the so-called MOS categories, see the UNIT DATA BASE DEFINITIONS section of the ACSDB Unit Data Base narrative. This paper also explains any estimation methodologies employed in generating the US non-divisional personnel data. It should be reviewed with other US Unit Data Base narratives for a comprehensive explanation of the personnel data recorded for US non-divisional units in the Unit Data Base. The paper is divided into three sections. The first is an introductory section explaining the scope of the task and the general approach used to address the task. The second discusses personnel data generation for US non-divisional combat units. The third discusses personnel data generation for US non-divisional service support units. Attachments at the end of the paper contain charts and tables providing data and information used to generate US non-divisional personnel data.

Personnel data of US non-divisional combat and service support units is recorded in the Unit Data Base of the ACSDB in records identified as:

-- (Army or Corps) Troops,
-- (Army or Corps) HQ, and
-- (ID, AD, or AbnD) Att.

- (Army or Corps) Troops: Units identified as such include the personnel of all non-divisional combat units (artillery, tank, tank destroyer, combat engineer, etc.) subordinated either to army or corps. For example, "XIX Corps Troops" is the designation assigned to the various non-divisional field artillery battalions, combat engineer battalions, tank destroyer battalions, etc., subordinated directly to US XIX Corps control. However, personnel data for non-divisional combat units subordinated to <u>divisions</u> under XIX Corps control is not recorded under "XIX Corps Troops," but rather under "(ID, AD, or AbnD) Att" as described below. The composition of the non-divisional combat units subordinated to armies and corps is described in the

"ATTACHMENTS & DETACHMENTS" section of the Unit Data Base, where the daily order of battle of combat units is recorded.

- (Army or Corps) HQ: Units identified as such include the personnel of all non-divisional service support units (medical, quartermaster, ordnance maintenance, etc.) subordinated either to army or corps. For example, "1st Army HQ" is the designation assigned to the various non-divisional quartermaster companies, ordnance maintenance companies, field hospitals, etc., subordinated to US First Army control. Since the composition of non-divisional service support units is not described in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base, a later section of this paper presents the US order of battle for non-divisional service support units.

- (ID, AD, or AbnD) Att: Units identified as such include the personnel of all non-divisional combat units subordinated directly to US infantry, armored, or airborne divisions. For example, "75th ID Att" designates the various non-divisional combat units (chemical mortar company, tank destroyer battalion, tank battalion, etc.) attached directly to the US 75th Infantry Division. These units almost invariably include the tank destroyer and tank battalions attached to US infantry divisions, which in practice became virtual organic divisional units by the time of the Ardennes Campaign. The composition of the non-divisional combat units subordinated to divisions is described in the "ATTACHMENTS & DETACHMENTS" section of the Unit Data Base, where the daily order of battle of combat units is recorded.

In generating personnel data of US non-divisional units for the ACSDB, two major considerations had to be addressed. First, as is the case with US divisional personnel data, there exists inconsistencies and contradictions in the historical personnel data of the units for which such data was found in research. For example, the daily on-hand personnel strength of a unit on one day does not always equal the unit's daily on-hand personnel strength of the previous day plus and minus the reported personnel gains and losses of the period between the two reports. Secondly, it proved impractical to obtain daily personnel data for the hundreds of US non-divisional units which participated in the Ardennes Campaign. Research in primary source records was directed at units of division-size and greater. Some divisional and corps records contain personnel data for non-divisional units, and this data is incorporated in the process used to generate non-divisional personnel data. However, the number of these units for which data was obtained is only a portion of the total number of US non-divisional units. Related to this problem was the question of how to record the data for the divisional units. To record the data of all non-divisional units individually, as was done with the divisions, would have created an accounting task and computer data base file of unmanageable dimensions, although it was recognized that this would have been

the most accurate methodology for recording the data.

To address these considerations, it was decided to record personnel data of non-divisional units in aggregate for "Army/Corps Troops," "Army/Corps HQ," and "ID/AD/AbnD Att" as described above. This approach utilizes personnel attrition and reinforcement (i.e., replacements and RTDs) rates, derived from the available historical data of non-divisional units which participated in the Ardennes Campaign. This approach was decided upon after considerable deliberation and only after all other possibilities had been evaluated. Chief among these alternatives was the utilization of the 12th Army Group G-1 Daily Summary, the same document discussed in the US Divisional Personnel narrative -- acquired from the US National Archives in Record Group 407, Box 1753, File 3, and Box 1754, File 4. Review of the 12th Army Group G-1 data showed that its applicability to the required task was not possible due to limitations in the data. For example, the 12th Army Group G-1s do not record daily casualties for US First and Third Army combat troops and the assigned strengths of these same units change only on a weekly basis. Strengths of various corps combat and service support troops frequently fluctuate on a daily basis, due to daily changes in corps orders of battle. Comparisons of these personnel strengths with the sum of the strengths of the subordinated non-divisional units -- the organization of which is recorded in the ATTACHMENTS & DETACHMENTS section of the Unit Data Base -- often shows some variance in the 12th AG G-1 strengths and the aggregate strengths of the subordinated units. Therefore, since the 12th AG G-1 data was inherently lacking in accuracy and a consistent correlation between it and the data derived from the order of battle was not possible, it was decided to base the personnel data on the order of battle. This decision was made in order to maintain consistency within the data base between the order of battle and personnel data, and to simplify accounting procedures, the necessity of which became most apparent when the scheme was actually implemented.

US NON-DIVISIONAL COMBAT UNIT PERSONNEL DATA

The daily order of battle recorded in the ATTACHMENTS & DETACHMENTS section of the Unit Data Base is used as a blueprint in estimating the personnel data of non-divisional combat units. Personnel strengths of the non-divisional combat units subordinated to an army or corps, or attached to a division, are summed by the various MOS personnel categories. (As explained in the UNIT DATA BASE DEFINITIONS section of the ACSDB Unit Data Base narrative, all personnel in non-divisional units are considered to belong to a single MOS category. i.e., all personnel in a 105mm howitzer towed field artillery battalion are "Artillery," and all personnel in a self-propelled tank destroyer battalion are "Armor." To the sum of these unit's personnel strengths are applied attrition and reinforcement rates derived from historical data. The calculated figures are the estimated personnel casualties and reinforcements by MOS category for the army troops, corps troops, or divisional attachments. The strengths, casualties, and reinforcements of the various MOS categories are totalled to provide the estimated total strengths, casualties, and reinforcements of the unit.

Attachment 1 shows the attrition and reinforcement rates applied in the above-described process for the following MOS categories: Armor, Infantry, Artillery, Engineer, and Other (antiaircraft artillery units). These rates are derived from the historical data of units listed in the attachment.

Examination of the daily on-hand personnel strengths of US army and corps troops and divisional attachments as recorded in the ACSDB shows fluctuations in the daily strengths caused by the attachment or detachment of units and consequent changes in the daily order of battle. It was recognized that a methodology was needed to account for personnel attrition sustained by non-divisional combat units, which would reduce a unit's personnel strength by subtracting its personnel losses when the unit moved from one parent organization to another. The key element of this methodology is the estimation of daily personnel strengths of non-divisional combat units, using the daily percentages of assigned strength to authorized strength of army and corps troops, percentages calculated using the 12th AG G-1 data. In the 12th AG G-1 data are found both assigned and authorized strengths of corps and army combat troops. The percentage of assigned to authorized is invariably less than 100 percent and often shows a recognizable decline during the period 16 December 1944-16 January 1945. When a non-divisional combat unit moves from one parent organization (army, corps, or division) to another, its strength on the day it is reassigned is estimated to be the same percent of its authorized strength as the percent of assigned to authorized of the new parent organization. Although

only a rough approximation, the approach was adopted partly to provide a means of compensating for some unit personnel attrition and, most importantly, to simplify what would have been an accounting process of unwieldy dimensions. The process used to estimate German non-divisional combat unit personnel strength does in fact utilize an approach in which strengths of all component units are estimated on a daily basis and then added together. This methodology is feasible for German non-divisional combat units because the daily German order of battle is much less comprehensive and detailed, and consequently has fewer attachments and detachments, than the US order of battle.

Along with the US 28th and 106th Infantry Divisions, and elements of other divisions, several non-divisional combat units suffered heavy personnel losses in the opening phases of the Ardennes Campaign. These units, their estimated personnel losses, and the derivation of their personnel attrition are provided in Attachment 2. Note that for the periods after these units had incurred their high personnel losses, they are assumed to be at the heavily attrited strengths when changing parent organizations in the ACSDB.

To simplify accounting of personnel attrition in US non-divisional combat units, the attrition rate applied to daily personnel strengths included battle and non-battle casualties aggregated. To break down these casualties into killed, wounded, captured/missing, and DNBI, a set of percentages was derived using the sums of the casualties presented in Attachment 1. The percentages, and the data used to calculate them are as follow:

	Killed	Wounded	Captured/Missing	DNBI
Armor	55	276	100	459
Infantry	15	28	13	63
Artillery	15	70	53	138
Other	6	23	1	101
TOTALS	91/1416	397/1416	167/1416	761/1416
Percents	6.43%	28.04%	11.79%	53.74%

US NON-DIVISIONAL SERVICE SUPPORT UNIT PERSONNEL DATA

The composition of US non-divisional service support units subordinated to corps and armies is not described in the ATTACHMENTS & DETACHMENTS section of the Unit Data Base where only combat units are listed. This composition is instead described in this paper along with the sources used in its preparation and methodologies and sources used to generate the personnel data of the US non-divisional service support units.

Attachment 3 is a description of the order of battle of the non-divisional service support units subordinated to US corps and armies. This order of battle is derived from a number of sources including:

- First United States Army. Report of Operations, 1 August 1944-22 February 1945. 4 vols.

- After Action Report, Third US Army: 1 August 1944-9 May 1945. Vol. 1, The Operations and Vol. 2, Staff Section Reports.

- Weekly 12th Army Group G-4 Reports Nos. 20-25, covering the period 10 December 1944 to 20 January 1945 (available in US National Archives Record Group 407, Box 1779, 99/12-4.1 to 99/12-5).

- V Corps G-1 After Action Reports for December 1944 and January 1945, including monthly station lists and strengths of command (available in US National Archives Record Group 407, Box 3412, 205-0.13 to 205-1.12).

- VII Corps Estimated Loss Reports (US National Archives Record Group 407, Box 3839, 207-1.8 to 207-1.16).

- XIII Corps Machine Records Unit (MRU), including 31 December 1944 list of assigned and attached troops (US National Archive Record Group 407, Box 4555, 213-0.3).

- XIX Corps G-1 Weekly Reports for the period 16 December 1944-20 January 1945 (US National Archives Record Group 407, Box 4983, 219-1).

- XX Corps G-1 Daily Summary for the period 16 December 1944-16 January 1945 (US National Archives Record Group 407, Box 5075, 220-1 to 220-1.2).

Using these sources and US Army Field Manual (FM) 101-10, Staff Officers' Field Manual: Organization, Technical and Logistical Data, 21 December 1944, it was possible to determine the "core" service support elements common to either armies or

corps, such as laundry companies, military police companies, band personnel, etc. These units are considered static, i.e., their assignment to army or corps does not change during the period 16 December 1944-16 January 1945. Their identification, personnel strengths, and assigned MOS categories, are shown in the first part of Attachment 3. Particularly useful in the determination of the "core" army and corps service support units was the XIX Corps Weekly Reports for the period 16 Dec 44-20 Jan 45, records fully referenced above. These documents provide the composition of corps troops for the US XIX Corps and the authorized and assigned strengths of its component non-divisional service support units.

Part 2 of Attachment 3 describes the order of battle changes for army and corps in medical, supply, ordnance, and transportation MOS categories. The order of battle changes mainly affect army units, although changes occur in corps, principally by the attachment or detachment of ordnance-maintenance units and field artillery/antiaircraft artillery/engineer group headquarters. The ordnance-maintenance unit changes for corps are listed in Part 2 of Attachment 3. Changes in the group headquarters are not listed. When changes are made in Corps HQ personnel strengths for the MOS categories of Artillery, Armor, Engineer, or Other, it is the result of the attachment or detachment of the personnel in the following units:

Antiaircraft Group HQ	73	Other personnel
Antiaircraft Brigade HQ	79	Other personnel
Armored Group HQ	98	Armor personnel
Field Artillery Group HQ	99	Artillery personnel
Field Artillery Brigade HQ	103	Artillery personnel
Tank Destroyer GroHQ H	72	Armor personnel
Engineer Group HQ	479	Engineer personnel

(An engineer group HQ normally controlled or included one light equipment company, one pontoon bridge company, and elements -- approximately one-half -- of one treadway bridge company. The personnel of these units are counted in the above figure.)

The orders of battle of non-divisional service support units for the US First and Third Armies are well documented in the sources listed above. Unfortunately, the Ninth Army's report of operations is not as informative as those of the First and Third Armies (see Conquer: The Story of the Ninth Army, 1944-1945, Washington, Infantry Journal Press, 1947). Therefore, it was necessary to estimate composition of certain service support units for the Ninth Army. This was achieved by comparing the average number of divisions or corps under Ninth Army control with comparable information for the First Army and estimating the number of service support units based on the ratio of divisions/corps between the two armies. Typically, the Ninth

Army was found to have about two-thirds of the divisional and corps assets of the First Army, so estimated numbers of service support elements were assigned accordingly to the Ninth Army.

In the same way that the daily order of battle of combat units serve as the blueprint for estimating "Corps Troops" and "Army Troops" personnel strengths, the organization of the non-divisional service support units as described in this paper is used as the blueprint for calculating the personnel strengths of the "Corps HQ" and "Army HQ." The 12th Army Group G-1 Daily Summaries contain aggregate daily personnel data for the service and headquarters troops of the armies and corps under the 12th Army Group. This data, however, is not broken down into the MOS categories and contains some of the same limitations as found in the data for the army and corps troops.

Personnel strengths for the army and corps HQs is based on the order of battle of non-divisional service support units. Estimation of personnel attrition and reinforcements is done using a different approach than that used for army and corps troops.

For armies (US First, Third, and Ninth Armies), the 12th AG G-1s serve as the main source of data for personnel losses and gains in non-divisional service support units. Daily battle and non-battle casualties are not provided in the 12th AG G-1s. However, cumulative casualties for the period 15 December 1944-16 January 1945 can be estimated by taking the difference of the cumulative casualty figures on those days. The differences are calculated as:

First	29 KIA	101 WIA	31 Cap/MIA	855 DNBI
Third	32	30	49	932
Ninth	18	55	1	846

These losses, particularly the battle casualties, are almost negligible. In all three cases the battle losses are less than the battle casualties of a single division on a day of heavy combat. Therefore, to facilitate accounting procedures and to compensate for the lack of daily casualty data, it was determined to assign losses at equal daily increments for the three armies.

For reinforcements, the only data available in the 12th AG G-1s for the three armies during the period 16 December-16 January is for the Third Army. On 22 and 30 December and 5 and 12 January respectively, reinforcements of 197, 271, 234, and 87 personnel (totalling 789) are reported. These are cumulative weekly figures. The same total (789) is used for the First Army, as its average daily personnel strength of non-divisional service support units and total casualties approximates that of the Third Army. For the Ninth Army, 75 percent of 789 (592 reinforcements) is estimated to be the total reinforcement figure for 16

December-16 January, a percentage based on the ratio of the average daily strengths of the Ninth and Third Armies' non-divisional service support units (approximately 36,000/48,000). Reinforcements are assigned to the armies in equal daily increments in the same manner losses are assessed against the armies. Reinforcements are broken down into replacements/returns to duty (RTDs) using a ratio taken from data for the US XIX Corps (XIX Corps G-1 Weekly Reports for the period 16 December 1944-20 January 1945 from US National Archives Record Group 407, Box 4983, 219-1). This ratio is derived from the total number of replacements/RTDs for five weekly reports between 16 December and 13 January (84 replacements/57 RTDs, i.e., 59.6%/40.4% -- these numbers are calculated using adjustments to the 23 December G-1 Report, which are made in light of the anomalies in the personnel data for the 817th Military Police Company on 16 and 23 December).

Casualties and replacements/RTDs are distributed proportionately among army non-divisional service support units using the ratios of the start strengths of the MOS categories to the start strength of the entire "Army HQ." For example, the First Army HQ's estimated strengths on 15 December are as follow:

- Armor	72
- Artillery	400
- Maintenance	13620
- Medical	10178
- Transportation	4742
- Supply	9816
- Engineer	5093
- Other	5540
TOTAL	49461

Casualties and reinforcements are distributed among the MOS categories by applying the following percentages to the total number of casualties and replacements/RTDs:

- Armor	0%	72/49461
- Artillery	1%	400/49461
- Maintenance	27%	13620/49461
- Medical	21%	10178/49461
- Transportation	10%	4742/49461
- Supply	20%	9816/49461
- Engineer	10%	5093/49461
- Other	11%	5540/49461

This scheme is used because there is no data on casualty and reinforcement distribution among army service support personnel. As discussed above, the battle casualties among these units are negligible. The majority of their losses are DNBI. Since there is no indication that any MOS category would experience a higher

proportion of DNBI losses, it is assumed that the losses would likely be incurred at an equal rate among the service support personnel, and that the distribution of reinforcements would follow a similar pattern as the units requested personnel to replace losses and received returnees from medical treatment.

When non-divisional service support units in armies are attached or detached, their personnel strengths are assumed to be at authorized levels. This is done primarily to simplify accounting procedures. In any event, as described in this paper, the changes in the non-divisional service support order of battle are relatively infrequent and the estimated losses sustained by the units very low.

Like the personnel data for armies, personnel data for non-divisional service support units subordinated to corps is generated primarily by using the data contained in the 12th AG G-1s. On-hand personnel strengths are calculated using the orders of battle presented in this paper as blueprints. Casualties and reinforcements are taken from the data of those corps HQs in the 12th AG G-1s with reliable data. Some corps HQs have no reinforcement data or casualty data. Some have casulaty date only, but this data cannot be used. See the 12th AG G-1 data on the XVIII Airborne Corps, for which no reinforcement data, on a daily basis or in weekly cumulatives, is recorded during the period 15 December-16 January. For the same period, see the 12th AG G-1 data on the VIII Corps's battle and non-battle casualty data. No daily casualty data is provided, and the cumulative casualties actually decrease between 1 and 16 January as the composition of the corps HQ changes.

The following data, extracted from the 12th AG G-1 Daily Summary, is used in generating corps HQ casualty and reinforcement data:

III Corps HQ	1	Battle Casualties	40	DNBI
V Corps HQ	49	"	101	"
XII Corps HQ	19	"	114	"
XIII Corps HQ	7	"	54	"
XX Corps HQ	11	"	103	"
XIX Corps HQ	12	"	120*	"

III Corps data is from daily reports for 15 Dec-16 Jan.
V Corps data is from cumulative reports for 15 Dec-16 Jan.
XII Corps data is from daily reports for 15 Dec-16 Jan.
XIII Corps data is from cumulative reports for 15 Dec-16 Jan.
XX Corps data is from daily reports for 15 Dec-16 Jan.

* Data from XIX Corps G-1 Weekly Reports for the period 16 December 1944-13 January 1945 (US National Archives Record Group 407, Box 4983, 219-1).

III Corps HQ	14 Reinforcements
XII Corps HQ	145 Reinforcements
XX Corps HQ	179 Reinforcements
XIX Corps HQ	141 Reinforcements*

III Corps data is from daily reports for 15 Dec-16 Jan.
XII Corps data is from daily reports for 15 Dec-16 Jan.
XX Corps data is from daily reports for 15 Dec-16 Jan.

* Data from XIX Corps G-1 Weekly Reports for the period 16 December 1944-13 January 1945 (US National Archives Record Group 407, Box 4983, 219-1).

This data is modified to compensate for the differences in average daily strengths of the units in the 12th AG G-1s and the start strengths of the units as estimated by the order of battle presented in this paper (the start strengths being approximations of the average daily strengths of the units as recorded in the ACSDB). The compositions of the corps HQs in the ACSDB differ in some cases from the 12th AG G-1 corps HQs, because the various corps evidently counted the personnel of some units with the corps combat troops. This may in part explain the differences between the ACSDB corps troops personnel strengths and those in the 12th AG G-1s. Thus, the data listed in the above tables is changed using the following ratios.

Unit	ACSDB Start Strength to Average Daily Strength in 12th AG G-1s	Ratio
III Corps	5581/2112	2.6/1
V Corps	6813/3072	2.2/1
XII Corps	7515/4558	1.6/1
XIII Corps	6916/4003	1.7/1
XIX Corps	6777/3901	1.7/1
XX Corps	7496/5898	1.3/1

Average daily strengths in the above table are determined by taking the aggregate of the daily strengths given in the 12th AG G-1s and dividing by the number of days, i.e.,

III Corps	69698/33	
V Corps	101392/33	
XII Corps	150427/33	
XIII Corps	112077/28	
XX Corps	194621/33	
XIX Corps	19505/5	(weekly strengths -- five reports)

Thus, the estimated casualties and reinforcements, as used in the ACSDB, for the following units are as follow:

	Casualties		Reinforcements	
III Corps:	107	(2.6x41)	36	(2.6x14)
V Corps:	329	(2.2x150)	--	
XII Corps:	213	(1.6x133)	232	(1.6x145)
XIII Corps:	104	(1.7x61)	--	
XX Corps:	149	(1.3x114)	233	(1.3x179)
XIX Corps:	225	(1.7x132)	240	(1.7x141)

These calculated figures are used to estimate casualties for the VII, VIII, and XVIII Corps and reinforcements for the V, VII, VIII, XIII, and XVIII Corps.

For casualty estimation, average daily battle casualty and DNBI rates are generated using the total 12th AG G-1 losses (and the XIX Corps G-1s for that unit) for the III, V, XII, XIII, XX, and XIX Corps and the average daily strengths of these units, or:

Total Strength	Battle Cas	DNBI	# Days
69698	1	40	33
101392	49	101	33
105427	19	114	33
112077	7	54	28
194621	11	103	33
19505	12	120	35 (5 weekly reports)
602720	99	532	165 TOTALS (incl. 5 weekly rpts)

602720/165 = 3653
99/165 = 0.51, 0.51/3653, or 0.01%/day -- Battle Casualties
532/165 = 2.7, 2.7/3652, or 0.07%/day -- DNBI

The rates of 0.01%/day and 0.07%/day are applied to the start strengths of the VII, VIII, and XVIII Corps, and the resulting figures are then multiplied by the number of days of participation by the unit to calculate the total number of estimated casualties of these three corps. These numbers are distributed evenly over the period of a corps' participation in the campaign. The battle casualties are broken down into killed, wounded, and captured/missing using the following percentages:

Unit	Killed	Wounded	Cap/MIA
V Corps	5	27	17
XII Corps	6	13	0
XIII Corps	1	6	0
XX Corps	2	9	0
TOTALS	14/86 (16.3%)	55/86 (64.0%)	17/86 (19.7%)

Casualty and reinforcement distribution are distributed proportionately among corps non-divisional service support units using the same methodology employed with army non-divisional

service support units, i.e., using the ratios of the start strengths of the MOS categories to the start strength of the entire "Corps HQ." For the US VIII Corps, additional casualties of 140 artillery personnel captured/missing on 17 December and 20 medical personnel captured/missing on 20 December are added to the casualties estimated using the above-described processes. These personnel represent the "Malmedy Massacre" involving troops of the 285th Field Artillery Observation Battalion on 17 December, and the capture of personnel of the 42d Field Hospital on 20 December, events well-documented in primary and secondary sources.

For the V, VII, VIII, XIII, and XVIII Corps, a ratio of reinforcements to casualties is used to estimate the reinforcements of these corps. This ratio is derived from the estimated reinforcements and casualties of the III, XII, XIX, and XX Corps.

III Corps	REINF 36	CAS	107	(0.336)
XII Corps	232		213	(1.089)
XIX Corps	240		225	(1.066)
XX Corps	233		149	(1.564)
				4.055

4.055/4 = 1.014, or 1.014 reinforcements per casualty

Reinforcements for all corps HQ units are broken down into replacements/RTDs using the same ratio of 59.6/40.4, as employed with the army HQ units. Replacements/RTDs are distributed evenly over the period of a unit's participation in the Ardennes and proportionately among MOS categories in the same manner as the casualties.

US Medical Facilities Data

Fields 38-42, and 59 in the Unit Data Base contain data for army-level medical facilities. This data is provided primarily for US units. It was not found in the historical record for German forces, so medical facilities data for German forces in the Ardennes Campaign Simulation Data Base (ACSDB) is limited to an estimate of the available "Number of Hospital Beds." The derivation of the German data is discussed in another narrative for the ACSDB final report.

The data provided for US Army medical statistics is recorded only for army-level units since the data found in research is compiled in aggregate for each of the three armies of the US 12th Army Group -- the First, Third, and Ninth Armies. US divisions in the Ardennes had elements of hospital units assigned to them in a command relationship most equivalent to "operational control" (OPCON) in the modern US Army. During World War II, hospital units generally served the personnel of a particular division or divisions. However, all reports on the operations were forwarded to the Army Surgeon's office. Thus, the reports at the army level include all of the statistics for hospital units under army command.

A report prepared by consultant Shelby L. Stanton for the ACSDB, titled Medical Evacuation and Hospitalization during the Ardennes Campaign, contains substantial information on the organization and activities of US Army medical units during the Ardennes. Relevant sections of the US First and Third Army After Action Reports also provide descriptions of the activities of the armies.

US medical facilities statistics are recorded in the Unit Data Base of the ACSDB in the records of the US army headquarters units, i.e., in the records identified with the names "1st Army HQ," 3d Army HQ," and "9th Army HQ."

The data entered in these records is taken directly from the sources listed below. All records are from US National Archives Record Group (RG) 112, Boxes 402 and 403, Office of the Surgeon General, World War II Administrative Records.

- Field 38 (FACILITIES & STATISTICS -- No. of Hospital Beds): Data for the First, Third, and Ninth Armies is taken directly from "HEADQUARTERS TWELFTH ARMY GROUP, Medical Section, Daily Status of Patients and Evacuation of Personnel, US Army, ending..."

- Field 39 (FACILITIES & STATISTICS -- No. of Beds Filled): Data is taken for the First, Third, and Ninth Armies from the same source used for Field 38 data.

- Field 40 (FACILITIES & STATISTICS -- No. of WIA Entering Hospital): Data for the First and Ninth Armies is taken from either "WEEKLY NUMBER OF ADMISSIONS AND DISPOSITIONS OF FIRST ARMY MEDICAL INSTALLATIONS SINCE D-DAY" (table dated 9 April 1945), or "WEEKLY NUMBER OF ADMISSIONS & DISPOSITIONS OF NINTH ARMY MEDICAL INSTALLATIONS SINCE 8 Sept 44" (table dated 9 April 1945). The data in these sources is weekly data and is divided by seven to estimate daily numbers of wounded admissions. Data for the Third Army is taken from an undated table titled "THIRD U.S. ARMY HOSPITALS DIRECT ADMISSIONS" which provides weekly data from 21 July 1944 to 2 February 1945. This data is also divided by seven to give estimated daily averages.

- Field 41 (FACILITIES & STATISTICS -- No. of DNBI Entering Hospital): Data for the three US armies is derived exactly as the data for Field 40.

- Field 42 (FACILITIES & STATISTICS -- No. Died in Hospital): Data for the three US armies is derived exactly as the data for Field 40.

- Field 59 (FACILITIES & STATISTICS -- No. Evacuated): Data for the Third and Ninth Armies is taken directly from "HEADQUARTERS TWELFTH ARMY GROUP, Medical Section, Daily Status of Patients and Evacuation of Personnel, US Army, ending..." This source is also used for First Army data on 15-16 December and 26 December-16 January. For 17-25 December the daily number of evacuated personnel from the First Army is estimated by taking the weekly number of admissions for the weeks ending 22 December and 29 December, as found in "NUMBER OF ADMISSIONS...SINCE D-DAY," and subtracting out the daily evacuated personnel figures for 16 December and 26-29 December. The resulting number is divided by nine to give an average daily number of admissions for the period 17-25 December. This estimation is performed because the source "NUMBER OF ADMISSIONS...SINCE D-DAY" does not report daily evacuated personnel figures during the period 17-25 December, omissions no doubt due to the confusion caused by the German offensive.

The math used in this estimation process is as follows:

((10791-1267) + (6873-930-871-970-1059))/9, or 1396 per day.

US AMMUNITION DATA

Attachment 1 is a list of the primary source records researched for the ACSDB and used to generate ammunition and other logistics data for US units in the Ardennes Campaign. Unless otherwise indicated, all data is from Record Group (RG) 407 in the US National Archives. Data from these sources is entered without modification (i.e., prorating) in the Unit Data Base if the data is recorded on a daily basis. If no daily primary source ammunition data is available in the records, various methodologies are employed to interpolated or extrapolate ammunition data on a daily basis.

For categorization of ammunition types as used for US Army units in the ACSDB, see the T/O&E Data Base narrative.

US Army Ammunition Supply.

The flow of ammunition supply described for US units in the ACSDB is as follows. All data is recorded for army- and corps-level units in the "troops" records, i.e., "1st Army Troops," "3d Army Troops," "VII Corps Troops,", "XVIII Corps Troops,' etc, with one exception as noted below.

- Army-level "on-hand" figures are estimated for the First and Third Armies to be three times the average daily consumption of the army, as armies by doctrine maintained a three-day supply of ammunition at all times. For the Ninth Army the amount on-hand is estimated to equal one day's average daily expenditure.

- Corps-level "on-hand" figures are estimated to be three to five times the average daily consumption of the corps, as corps by doctrine maintained a three- to five-day supply of ammunition at all times. Note that for "corps HQ" units ("XVIII Corps HQ," "XX Corps HQ," etc.), on-hand supply for "Tank/AT" and "Other" ammunition is recorded in the ACSDB. These figures represent the ammunition levels maintained by the organic army and corps service support units. This is the exception noted above.

- Division-level "on-hand" figures are assumed to be the total doctrinal ammunition loads as listed in FM 101-10, Staff officers' Field Manual: Organization, Technical and Logistical Data, 21 December 1944 when actual historical data is not available in records. Total ammunition includes the basic load, and that which was carried in the unit trains. In the ACSDB, US on-hand ammunition figures are reduced only when heavy ammunition expenditures cause the amount of ammunition consumed to exceed the maximum resupply capacity in tons per day, or when unit supply routes are known to have been interrupted.

- Army-, corps-, and division-level "received" ammunition figures are considered to equal ammunition consumed by the unit unless the amount consumed exceeds the transport capacity of a unit, or unless enemy action interdicted supply lines.

- Army-level "used" figures include the amount of ammunition transferred to, and received by, the corps in the army, as well as the amount of ammunition that was expended by organic army assets.

- Corps-level "used" figures include the amount of ammunition transferred to and received by the divisions in the corps as well as the amount of ammunition that was expended by organic corps assets, including non-divisional units attached to divisions.

- Division-level "used" figures include only the amount of ammunition expended by organic divisional units of the division.

An exception to the above-described flow of supplies is the aerial resupply of the 101st Airborne Division. Amounts of ammunition received by the 101st during aerial resupply operations from 21-29 December 1944 are not included in the VIII Corps Troops ammunition consumption (used) totals since these supplies originated in the communications zone.

US Ammunition Data Estimation.

When sufficient data is available in the primary source records, it is used to generate data for the above-described requirements. However, estimation is required for some units. Examples of the major estimation methodologies used to fill in missing data points include the following.

- US First Army and US Third Army records (the armies' "reports of operations") contain excellent ammunition supply records from which corps data is in some cases estimated. Ammunition data for days on which individual corps data is unavailable is derived from army-level records of ammunition consumed and ammunition received by taking the aggregate army figures and subtracting from them the known figures for corps and distributing the remainder among the corps for which data is not available. This same approach for divisions is used in cases where corps ammunition data is known but ammunition figures are not available for division(s). Adjustment of some values derived from the estimation process insures that no unit expends more ammunition than the amount for which it has transport capacity.

- US Ninth Army ammunition data is estimated by creating ratios of the number of divisions in the First and Ninth Armies and then using these ratios to calculate Ninth Army ammunition statistics from First Army figures.

- Ammunition data from records compiled on a weekly or monthly basis are prorated by dividing the aggregate data by the total number of days of the period.

- For the US 2d Armored Division, ammunition data for the US 3d Armored Division is used to estimate the ammunition data. Both units were organized as heavy armored divisions, larger than the standard armored division organizations.

- For all units not engaged in combat or in reserve, the amount of ammunition used is assumed to be none, unless the historical record indicates that ammunition was consumed.

US FUEL DATA

The organization of fuel supply for US units as recorded in the ACSDB is identical to that used for ammunition supply for US units except for the inclusion of attached units with divisions. For US infantry, airborne, and armored divisions, the fuel data of the "standard" attached combat battalions is added to the fuel data for the division organic units. In the case of the infantry and airborne divisions these include one tank battalion, one tank destroyer battalion, and one anitiaircraft artillery battalion. In the case of the armored division, these include one tank destroyer battalion and one antiaircraft artillery battalion.

Fuel data is also calculated for army HQ and corps HQ units in the ACSDB, representing the fuel supply of the service support units maintained by army and corps.

US Fuel Data Estimation.

When fuel data for US units is available in primary sources, it is used in the ACSDB. The US First Army and Third Army After Action Reports contain daily fuel consumption (used) figures for the two armies, as do some divisional records.

On-hand fuel data for army and corps is based on three times the average amount of fuel consumed per day to represent the three days of Class III supply maintained by armies and corps. Note that this data, and the fuel received and fuel used data as well, includes all Class III supply -- petroleum, oil, and lubricants -- and not just motor fuel. This is because the data in the historical records for these units includes all POLs. For divisions, the fuel data (on-hand, received, and used) includes only motor fuel.

Fuel used by army troops includes fuel consumed by separate combat units of the army plus fuel supplied to subordinate corps of the army. The formula used to calculate consumption by the separate combat units is calculated using an equation from FM 101-10, December 1944, Chapter 3, Page 13:

MD = Movement Distance
SD = Supply Distance
UD = Unit Distance

MD + (.2 x SD) + 10 = UD (Administrative consumption of fuel is a variable value, in this case it equals 10.)

UD x 109.2 = Gallons Consumed (The amount of fuel required to move all vehicles of army troops units one mile is 109.2 gallons.)

US OTHER SUPPLY DATA

As described in the ACSDB T/O&E Data Base narrative, "Other Supply" weight is estimated using data for Classes I, II, and IV supplies as found on page 308 of FM 101-10, 21 December 1944. Other Supply data for US units in the Unit Data Base is estimated using the data derived from FM 101-10, 21 December 1944, for the US T/O&Es.

The estimated T/O&E Other Supply weights of divisions is used to calculate the on-hand Other Supply weights for infantry, armor, and airborne divisions in the unit data base. For "Corps HQ," "Corps Troops," "Army HQ," and "Army Troops," the on-hand personnel strength is multiplied by 24.99 pounds (the pounds of "Other Supply" used per man per day, as taken from FM 101-10, 21 December 1944), to estimate the on-hand weight of "Other Supply."

Other Supply received is equal to Other Supply used in all but a few special circumstances in the US Army in the Ardennes Campaign. Exceptions include the 28th and 106th Infantry Divisions, the 101st Airborne Division, and the 7th Armored Division, units whose supply was disrupted during the initial German attacks of the Ardennes Campaign. In these cases, the replacement of used Other Supply did not occur on the days on which those units were isolated.

Other Supply Used is calculated by multiplication of the pounds per man per day figure described above by the number of on-hand personnel in a formation on a given day. In cases where units were out of supply, rationing is represented in the ACSDB by distributing the amount of Other Supply on-hand across the period of no supply until the unit surrendered or regained a source of supply.

US TRANSPORTATION CAPACITY

Long-haul transport capacity for US army and corps is recorded in the Unit Data Base in records of the "army HQ" and "corps HQ." The capacity is based on a cargo capacity of 175 tons per quartermaster truck company, the principal long-haul transport unit in the US Army during World War II, and the number of quartermaster truck companies per army or corps. This capacity includes that of the company's 2-1/2 ton trucks and 1 ton trailers.

Transportation Capacity (Wet) is estimated to be 10.1% of the total transport capacity of the long-haul transport assigned to the unit. The percentage is derived from US First Army figures for the period 16 December 1944-22 February 1945 as found in First United States Army: Report of Operations, 1 August 1944-22 February 1945, Vol. 4, page 122 (47,730 tons of Class V supplies out of 471,761 tons of all Classes of supplies). The number of gallons of 80 octane gasoline per ton is 326. This figure comes from page 214 of FM 101-10, Staff Officers' Field Manual: Organization Technical, and Logistical Data, July 1953. (The July 1953 version of this manual is used because the 21 December 1944 version does not have a gasoline weight.)

The following tables show the values used for long-haul transport capacity in the ACSDB.

Unit	Period	Total Tons	Fuel (1000 Gals.) (x.101)	Gals.)	Dry (Tons)
III Corps	All days	350	35.35	11.524	314.65
V Corps	All days	350	35.35	11.524	314.65
VII Corps	All days	350	35.35	11.524	314.65
VIII Corps	All days	350	35.35	11.524	314.65
XII Corps	All days	350	35.35	11.524	314.65
XIII Corps	All days	350	35.35	11.524	314.65
XVIII Corps	28 Dec-16 Jan	350	35.35	11.524	314.65
XIX Corps	All days	350	35.35	11.524	314.65
XX Corps	All days	350	35.35	11.524	314.65

Unit	Period	Total Tons	Fuel (1000 Gals.) (x.101)	Gals.)	Dry (Tons)
1st Army	15-16 Dec	6475	653.975	213.196	5821.025
	17-29 Dec	7175	724.675	236.244	6450.325
	30 Dec-5 Jan	7175	724.675	236.244	6450.325
	6-15 Jan	7175	724.675	236.244	6450.325
3d Army	15-16 Dec	4375	441.875	144.051	3933.125
	17-29 Dec	4375	441.875	144.051	3933.125
	30 Dec-5 Jan	4550	459.55	149.813	4090.450
	6-15 Jan	5250	530.25	172.862	4719.750

Derivation of the US Order of Battle

INTRODUCTION

Fields 60-79 of the Unit Data Base are used to describe the daily order of battle of forces tracked in the Ardennes Campaign Simulation Data Base (ACSDB). For complete information on the methods used to record the order of battle, see the section titled "ATTACHMENTS & DETACHMENTS -- TYPE, UNIT NAME" of the ACSDB Unit Data Base narrative.

The task of compiling the daily US Army Order of Battle for the ACSDB was aided by abundant and readily available primary and secondary sources of information. This paper is intended to serve as a bibliographic reference for the most commonly used sources. In some cases, particularly with non-divisional units, the information found in sources is fragmentary and sometimes contradictory. Therefore, this paper also serves as a basic description of some of the problems encountered in preparing the US Order of Battle, steps taken to address these problems, and the nature of the US Army's Order of Battle organizational scheme as used in World War II.

The last section of this paper lists the most important sources used in the compilation of the daily US Order of Battle, as well as the abbreviated references for them used in the text of this paper.

US Army doctrine in World War II normally used two descriptions for a unit's subordination status. These were:

1) Assigned
2) Attached

These descriptions could be and were used together to give a precise description of a unit's status. For example, the US 7th Armored Division on 17 December 1944 was assigned to the US Ninth Army but was attached to the US VIII Corps of the US First Army. Thus, whenever known, effective dates of change of attachment are reflected in the US Order of Battle rather than dates of change of assignment, as changes of assignment in many instances occurred days or even weeks after the unit was under the operational control of (i.e., attached to) another headquarters.

US non-divisional artillery units, most commonly corps artillery battalions, are often noted in sources as "supporting" a specific unit, usually a division. In these cases, if the artillery unit is not found in any source to have been actually attached to the unit, the artillery is not noted as being attached in the US Order of Battle.

BRITISH PERSONNEL DATA

British Personnel Start Strengths (20 December 1944).

Personnel start strengths of British units were estimated by using the personnel strengths of units found in SHAEF G-1, 14 December, and subtracting from them the battle casualties and disease and non-battle injury (DNBI) losses for the period between 14 December and the date when the British unit was first recorded in the Ardennes Campaign Simulation Data Base (ACSDB), usually 20 December. The strength of the 6th Airborne Division was not listed in the SHAEF G-1, 14 December. Its strength was estimated from its average battalion strength in RG331, 704-1.2, when it first appeared in the ACSDB. Personnel strengths of non-divisional combat units were estimated to equal the relevant authorized T/O&E strengths. This estimation was necessary as all personnel strengths for these units were aggregated under corps troops in SHAEF G-1, 14 December. (A copy of SHAEF G-1, 14 December, is found in Attachment 2 of this paper. It was obtained from the US National Archives.) British XXX Corps non-divisional supply personnel start strengths were taken from WO171/354. All other service support and corps headquarters personnel start strengths were estimated from contemporary British VIII Corps figures (WO171/292), prorated to reflect the difference in the reported VIII and XXX Corps HQ strengths given in SHAEF G-1, 14 December.

British Daily Personnel Strengths.

Daily British personnel strengths of divisions/brigades were determined by matching the average battalion strengths for 14 December of division/brigades as given in RG331, 704-1.2, with the SHAEF G-1, 14 December strengths of the units. (RG331, 704-1.2, has daily average battalion strengths of British units for the period of the Ardennes Campaign.) This procedure gave a ratio of total unit personnel strength to average battalion strength of each division/brigade which was used as a multiplier with the average battalion strengths from RG331, 704-1.2, to estimate daily division/brigade personnel strengths.

British Daily Personnel Battle Casualties.

Daily British personnel battle casualties for divisions/brigades were taken from WO171/348, WO171/4084, and RG331, 704-1.2. In the few cases where reports conflicted, casualties in RG331, 704-1.2, were taken as definitive as these reports were assembled later and were assumed to be corrected versions.

British Daily Replacements/Returns to Duty (RTDs)/DNBI.

This personnel data was estimated by subtracting from the daily personnel strength the following day's on-hand personnel strength, and from the resultant number was subtracted the current day's battle casualties. The remainder, either "+△" or "-△," was assumed to be generated by either Replacements/RTDs in the case of a "-△," or DNBIs in the case of a "+△".

Miscellaneous Information on British Personnel Data.

Due to the lack of data in sources and time restrictions in the ACSDB schedule, certain personnel data requirements were not met for British units. If necessary for analytical purposes, missing data may be generated using rates of comparable US units. The following descriptions apply:

- No daily military occupational specialty (MOS) data is provided for British brigades and divisions in the ACSDB.

- Battle casualties, DNBI, and replacements/RTDs are not provided for divisional attachments ("53d ID Att," "Guards AD Att," etc.). Only on-hand personnel strengths are provided for these units.

- Battle casualties and replacements/RTDs are not provided for the "XXX Corps HQ."

- Differentiation between replacements and RTDs is not made for any British unit in the ACSDB. All reinforcement figures are recorded as "REPLACEMENT" personnel in the Unit Data Base and should be considered reinforcement personnel, composed of both new replacements and returnees from medical treatment.

company, the cargo capacity of the trailers and trucks, and the number of companies in the corps. The breakdown into "Transportation Capacity (dry)/Transportation Capacity (wet)" used the same ratio used for the US units, i.e., 10.1 percent of total capacity was estimated to be for wet transport capacity.

Description of British Supply Flow in the ACSDB.

Data recorded in the ACSDB for British logistics is intended to reflect a flow of supplies similar to that used for the US and German units in the ACSDB.

- "XXX Corps HQ" supply amounts on-hand include the ammunition and fuel with service support units and supplies kept in corps dumps. "XXX Corps HQ" supply expended includes all supplies used by service support units and supplied to subordinated divisions, corps non-divisional combat units, and divisional attachments. "XXX Corps HQ" supply received is estimated to equal the amount of supply required to maintain the previous day's "XXX Corps HQ" on-hand supply amounts or the on-hand supply amounts recorded in sources.

- "XXX Corps Troops" supply amounts on-hand include the ammunition and fuel with non-divisional combat units attached directly to corps. "XXX Corps Troops" supply amounts expended and used are the amounts for the same units.

- Ammunition and fuel data for divisional attachments is recorded with the appropriately identified unit ("Guards AD Att," "53d ID Att," etc.).

- Note that the large increase of fuel and other supplies shown on 30 December reflects the opening of the XXX Corps supply installations on that date. Prior to 30 December all XXX Corps units drew supplies directly from British Second Army railheads.

- Note that the XXX Corps HQ consumption figures for artillery ammunition on 14-16 January reflects the relocation of the corps stocks to the British Second Army area.